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Faculty of Management

Descriptive analytics

Third task set report
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1 Task 3.1

1.1 Description

The aim of the first task was to verify the adequacy and statistical reliability of the specified autoregressive model:

$$y_t = \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 y_{t-7} + \epsilon$$

by applying a set of diagnostic tests at the 1% significance α level. The verification process focused on determining whether the estimated model satisfies the fundamental assumptions required for a valid time series regression. In particular, the analysis requires examining the autocorrelation structure of the residuals for the first ten lags in order to identify any remaining serial dependence that would indicate unrecognized dynamics. Furthermore, the task involved evaluating the presence of collinearity among the explanatory lagged variables since strong linear relationships between regressors may distort parameter estimates, increase standard errors, and reduce the interpretability of the model.

1.2 Autocorrelation check for 10 first lags

```
maxlag = 10;
alpha = 0.01;
a = autocorr(e, 'NumLags', maxlag);
a = a(2:end);
% H0 : autocorrelation coeffs until lag maxlag are 0
Q = N*(N+2)*sum(a.^2 ./ (N-(1:maxlag)))';
p_val_q = 1 - chi2cdf(Q, maxlag);
if p_val_q < alpha
    disp('Autocorrelation present in data')
else
    disp('No autocorrelation present in data')
end
% LM | Auxillary Regression
y_aux = e(maxlag+1:end);
E = zeros(length(y_aux), maxlag);
for i = 1:maxlag
    E(:,i) = e(maxlag+1-i:end-i);
end
X_aux = [X(maxlag+1:end,:), E]; % matrix for the auxiliary regression
betas_aux = OLS(X_aux, y_aux);
e_aux = y_aux - X_aux*betas_aux;
r2_lm = 1 - var(e_aux)/var(y_aux);
% test statistics
LM = r2_lm*length(e_aux);
p_val_lm = 1 - chi2cdf(LM, maxlag);
if p_val_lm < alpha
    disp('Autocorrelation present in data')
else
    disp('No autocorrelation present in data')
end
```

Listing 1: Checking autocorrelation for 10 first lags

The results of both the Q-test and the LM test indicate that there is no statistically significant autocorrelation present in the residuals for the first ten lags. Since the obtained p-values in both tests were greater than the adopted significance level of $\alpha = 1\%$, the null hypothesis of no autocorrelation cannot be rejected. This suggests that the residuals behave as white noise and that the model successfully captures the dynamic structure of the analyzed time series without leaving systematic serial dependence in the error term.

1.3 Collinearity check

```
X_t = X - mean(X);
VIF = zeros(1, K);
for k = 1:K
    idx = 1:K;
    idx(k) = [];
    y_aux_vif = X_t(:,k);
    x_aux_vif = X_t(:,idx);
    betas_vif = OLS(x_aux_vif, y_aux_vif);
    e_aux_vif = y_aux_vif - x_aux_vif*betas_vif;
    r2_vif = 1 - var(e_aux_vif)/var(y_aux_vif);
    VIF(k) = 1/(1-r2_vif);
    fprintf('VIF for Y[t-%d]: %.4f\n', k, VIF(k));
end
% eigenvalues of the matrix
eigs = eig(X_t'*X_t); % 0.0329; 0.2435; 1.3094
CI = sqrt(max(eigs)./eigs); % 6.3084; 2.3189; 1.0000
ConditionNumber = max(CI);
% VIF interpretation
if any(VIF > 10)
    disp('VIF number is greater than 10, thus multicollinearity exists in data')
else
    disp('VIF number is lower than 10, thus multicollinearity does not exist / is weak')
end
% CI interpretation
if ConditionNumber > 30
    disp('CI bigger greater than 30, thus multicollinearity exists in data')
elseif ConditionNumber > 15
    disp('CI between 15 and 30, thus multicollinearity is somewhat strong')
else
    disp('CI lower than 15, thus multicollinearity does not exist / is weak')
end
```

Listing 2: Collinearity check using VIF and CI

The collinearity check indicates that collinearity is not a serious problem in the estimated autoregressive model. The calculated VIF values for Y_{t-1} , Y_{t-2} and Y_{t-7} are 8.2761, 8.5243, and 1.7176 respectively, all of which remain below the critical threshold value of 10. Although the first two lagged variables exhibit relatively higher VIF values, they still suggest only moderate correlation rather than collinearity. This conclusion is additionally confirmed by the CI, which is lower than 15, indicating weak or nonexistent collinearity among the explanatory variables. Therefore, the estimated coefficients can be considered sufficiently stable and reliable for interpretation, and the model does not suffer from significant distortions caused by strong linear dependence between regressors.

2 Task 3.2

2.1 Description

The aim of the second task was to verify the adequacy of the following regression model:

$$Price_t = \beta_1 Price_{t-1} + \beta_2 Consumption_t + \beta_3 Consumption_{t-1} + \epsilon$$

The verification procedure was intended to assess whether the model satisfies the key assumptions required for reliable estimation and interpretation of the relationships between price and consumption variables over time. In particular, the task required examining the homoscedasticity assumption, which determines whether the variance of the error term remains constant across observations, since the presence of heteroscedasticity may lead to inefficient estimates and invalid statistical inference. Additionally, the stability of the estimated parameters had to be evaluated in order to determine whether the coefficients remain consistent throughout the analyzed period.

2.2 Homoscedasticity check

```
% we will use both White and Breusch-Pagan LM tests
% (both of them are also viable method of verification)
% White-test
x_white = X;
for i = 1:size(X,2)
    for j = i:size(X,2)
        x_white = [x_white, X(:,i).*X(:,j)];
    end
end
x_white(:, end+1) = 1;
betas_2 = OLS(x_white, e.^2);
e2 = e.^2 - x_white*betas_2; % auxiliary model errors
r2 = 1 - var(e2)/var(e.^2);
stat_w = r2*N;
m = OriginalK + OriginalK*(OriginalK+1)/2;
pval_white = 1 - chi2cdf(stat_w, m);
% Breusch-Pagan test
g = e.^2 / (e'*e/N) - 1;
Z = X;
stat_bp = 0.5 * g' * Z * inv(Z'*Z) * Z' * g;
pval_bp = 1 - chi2cdf(stat_bp, OriginalK);
```

Listing 3: Homoscedasticity using Breusch-Pagan and White tests

```
WHITE: We cannot reject the null hypothesis, thus proving the residuals are
        homoscedastic
BREUSCH-PAGAN: We cannot reject the null hypothesis, thus proving the residuals
        are homoscedastic
```

Listing 4: Results interpretation of White and Breusch-Pagan tests

The results of both the White test and the Breusch–Pagan test indicate that the assumption of homoscedasticity is satisfied in the estimated model. In both cases, the null hypothesis of constant variance of the residuals could not be rejected at the adopted significance level, which implies that there is no statistical evidence of heteroscedasticity. Consequently, the variance of the error term can be considered stable across observations, confirming that the residuals are homoscedastic.

2.3 Parameters stability check - CUSUM test



Figure 1: Parameters X and Y stability check

The CUSUM test results suggest that the estimated model parameters remain generally stable throughout the analyzed period. The cumulative sum statistic initially decreases from the zero level until approximately the 300th iteration and afterwards begins to increase, eventually approaching the lower critical boundary at the value of -20 . However, since the CUSUM statistic does not cross the critical lines, the null hypothesis of parameter stability cannot be rejected. This indicates that there is no strong statistical evidence of structural instability or significant changes in the coefficients over iterations. Although the movement of the statistic toward the lower boundary may suggest the presence of moderate fluctuations or temporary shifts in the model dynamics, these changes are not sufficiently large to undermine the overall stability and reliability of the estimated model. Therefore, the regression coefficients can be considered relatively consistent across the examined sample period, supporting the validity of the model for inference and prediction.

3 Task 3.3 - 5 points variant

3.1 Description

The aim of the third task was to estimate, optimize, and comprehensively verify a wage determination model for individuals of the same sex and birth month as mine, using the following specification:

$$wage = \beta_1 Experience + \beta_2 Skills + \beta_3 Yearsincompany + \beta_4 Supervisor'sassessment + \beta_5 Motivation + \epsilon$$

The task required performing a detailed econometric analysis to evaluate whether the model satisfies the classical assumptions of linear regression and whether the estimated relationships between wage and the explanatory variables are statistically reliable. In particular, the verification procedure included testing the autocorrelation of residuals for the first twenty lags using the LM test in order to determine whether serial dependence remains in the error structure, as well as examining homoscedasticity through the White test to assess whether the variance of residuals is constant across observations. The assignment further required analyzing multicollinearity among explanatory variables by calculating VIFs, identifying potentially correlated regressors whenever evidence of collinearity exists, and discussing the consequences of such relationships for the precision and stability of the estimated coefficients. Additionally, the task involved determining the maximum significance level α for which the residuals may still be considered both homoscedastic and free from autocorrelation, thereby evaluating the overall robustness of the model assumptions. A substantial part of the analysis was devoted to selecting an optimal specification through forward stepwise regression, beginning with an empty model and sequentially introducing variables according to the statistical significance of their associated t-tests until no additional variable satisfies the 5% significance criterion. Each stage of the variable selection procedure must have been clearly documented and interpreted in order to justify the final specification. After obtaining the optimized model, an additional verification process must have been conducted, including testing residual autocorrelation using the Q-test, examining homoscedasticity with the Breusch–Pagan test, and evaluating multicollinearity through the conditional number criterion.

3.2 Autocorrelation, Collinearity & Homoscedasticity check

```
p_lags = 20;
E_lag = zeros(N - p_lags, p_lags);
for i = 1:p_lags
    E_lag(:, i) = e_init(p_lags - i + 1 : end - i);
end
X_lm = [X(p_lags+1:end, :), E_lag];
e_curr = e_init(p_lags+1:end);
beta_lm = (X_lm'*X_lm)^-1 * X_lm' * e_curr;
e_lm_aux = e_curr - X_lm*beta_lm;
R2_lm = 1 - (e_lm_aux'*e_lm_aux) / (e_curr'*e_curr);
stat_lm_autocorr = (N - p_lags) * R2_lm;
pval_autocorr_lm = 1 - chi2cdf(stat_lm_autocorr, p_lags);
% homoscedasticity using White test
x_white = X;
for i = 1:OriginalK_2
    for j = i:OriginalK_2
        x_white = [x_white, X(:,i).*X(:,j)];
    end
end
x_white(:, end+1) = 1;
betas_white = OLS(x_white, e_init.^2);
e2_white = e_init.^2 - x_white*betas_white;
r2_white = 1 - var(e2_white)/var(e_init.^2);
stat_w = r2_white * N;
m_white = OriginalK_2 + OriginalK_2*(OriginalK_2+1)/2;
pval_white = 1 - chi2cdf(stat_w, m_white);
```

Listing 5: Autocorrelation and homoscedasticity check using White and LM tests

```

    % VIF collinearity
VIFs = zeros(1, OriginalK_2);
for i = 1:OriginalK_2
    y_vif = X(:, i);
    X_vif = X;
    X_vif(:, i) = [];
    beta_vif = (X_vif'*X_vif)^-1 * X_vif' * y_vif;
    e_vif = y_vif - X_vif*beta_vif;
    R2_vif = 1 - (e_vif'*e_vif) / (y_vif'*y_vif);
    VIFs(i) = 1 / (1 - R2_vif);
end

```

Listing 6: Collinearity check using VIFs

The diagnostic results indicate that the estimated model satisfies the assumption of homoscedasticity but suffers from collinearity among the variables. According to the White test's result, the null hypothesis of constant variance of residuals cannot be rejected, which confirms that the residuals are homoscedastic and that the variance of the error remains stable across the model. This supports the validity of the estimated standard errors and statistical inference. However, the calculated VIF values exceed the critical threshold of 10, indicating the presence of significant collinearity in the dataset. In particular, the variables “Years in Company” and “Experience” appear to be strongly correlated, which is economically intuitive because employees with greater overall work experience often tend to spend more years within the same company.

3.3 Forward stepwise regression

```

% forward stepwise regression
included_vars = [];
available_vars = 1:OriginalK_2;
while true
    best_pval = 1;
    best_var = 0;
    for i = available_vars
        cols_to_test = [included_vars, i];
        X_test = X(:, cols_to_test);
        [beta_test, ~, se_test] = OLS(X_test, Y);
        % t-test with pval
        t_stat = beta_test(end) / se_test(end);
        df = N - length(cols_to_test);
        pval = 2 * (1 - tcdf(abs(t_stat), df));
        if pval < best_pval
            best_pval = pval;
            best_var = i;
        end
    end
    if best_pval < 0.05 % threshold was specified to be 5%
        included_vars = [included_vars, best_var];
        available_vars(available_vars == best_var) = [];
        fprintf('Added variable %d (%s) with p-value: %.4f\n', best_var,
            variable_names{best_var}, best_pval);
    else
        break;
    end
end
end

```

Listing 7: Forward stepwise regression build

The forward stepwise regression procedure identified “Experience” and “Skills” as the most statistically significant explanatory variables for the wage model. Starting from the empty specification, the variable “Experience” was added first because it produced the lowest p-value - equal to 0, indicating a very strong relationship with the dependent variable. In the next iteration, “Skills” was included in the model, also with a p-value of 0. Since both variables are statistically significant at the 5% significance level, their inclusion improves the overall quality and predictive capability of the regression model. The results suggest that professional experience and employee skills are the most important factors influencing wages in the analyzed sample, while the remaining variables did not provide additional statistically significant explanatory information during the selection process.

3.4 Comparison - old vs newly obtained model verification

```
"OLD" MODEL VERIFICATION
Autocorrelation p-val -> 0.4890
Homoscedasticity p-val -> 0.0108
VIFs ->
    167.0749    7.4344    167.8798    4.3250    4.0696

Maximum alpha value for homoscedastic & autocorrelated residuals -> 0.0108
VIF number is greater than 10, thus multicollinearity exists in data
```

Listing 8: Old model verification results

The verification results for the original regression model indicate that the model satisfies the assumption of no autocorrelation but exhibits problems related to heteroscedasticity and collinearity. The autocorrelation test produced a p-value of 0.4890 value, which is substantially higher than conventional significance levels, meaning that the null hypothesis of no autocorrelation cannot be rejected and the residuals may be considered independent over time. However, the homoscedasticity test returned a p-value of 0.0108, implying that homoscedasticity can only be accepted for significance levels below 1.08%, making the model sensitive to stricter statistical criteria. Additionally, the calculated VIF values reveal severe collinearity, particularly for two explanatory variables with VIFs exceeding 167. Such extremely high values indicate strong linear dependence among regressors, which may substantially distort coefficient estimates, deepen the problem regarding standard errors, and reduce the stability of the model. Consequently, despite the absence of autocorrelation, the original specification cannot be considered fully reliable due to the significant collinearity problem and the relatively weak strength of the homoscedasticity assumption.

```
NEWLY OBTAINED MODEL VERIFICATION
Variables included ->      1      2

Autocorrelation p-val -> 0.3317
Homoscedasticity p-val -> 0.2415
CI -> 1.8396
Maximum alpha value for homoscedastic & autocorrelated residuals -> 0.2415
CI lower than 15, thus multicollinearity does not exist / is weak
```

Listing 9: Newly obtained model verification results

The verification results for the newly obtained model demonstrate a substantial improvement in overall econometric quality compared to the original specification. The autocorrelation test produced a p-value of 0.3317, indicating that the null hypothesis of no autocorrelation cannot be rejected and confirming that the residuals behave independently. The homoscedasticity test resulted in a p-value of 0.2415, which provides strong evidence that the variance of residuals remains constant across observations and suggests that the model is statistically robust even at relatively high significance levels. Furthermore, the CI value of 1.8396 is considerably below the critical threshold of 15, indicating that collinearity is either nonexistent or very weak in the optimized specification. These results imply that the forward stepwise regression procedure successfully eliminated problematic variables and produced a more stable and reliable model with stronger statistical properties and more trustworthy coefficient estimates.